To begin a thorough discussion of multivariable calculus, we need to first get comfortable graphing in three dimensions. We are familiar with the 2-dimensional xy-plane, but we now introduce a third variable: z. While there are many ways to do this, we traditionally build the following picture, where equations like $y = 2$ now build a *plane* instead of a line:

Example. Interpret the following equations and inequalities geometrically:

(i) $x \geqslant 0, y = 0, z \geqslant 0$

(ii) $z = x^2$

(iii) $z \geqslant x^2, -2 \leqslant y \leqslant 0$

(iv) $x = 3, y = 7$

Example. Compute the distance between $P(-3, -5, 1)$ and the yz-plane.

More generally, the distance between two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is given by

$$d(P_1, P_2) = |P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Example. Compute the distance between $P_1(1, 3, 4)$ and $P_2(5, -1, 6)$.

Definition A *sphere* is a set of points (x, y, z) that are all the same distance, r, from a fixed point, (x_0, y_0, z_0), called the center:

Example. Find the center and raidus of the following sphere:

$$x^2 + y^2 + z^2 - 2x - 6z = 0.$$

Example. Interpret the following equations and inequalities geometrically:

(i) $x^2 + y^2 + (x + 11)^2 = 100, z = -3$

(ii) $x^2 + y^2 = 25, z = x$

(iii) $1 \leqslant x^2 + y^2 + z^2 \leqslant 9$

(iv) $x^2 + y^2 + z^2 \leqslant 9, z \geqslant 0$

(v) $x^2 + y^2 + z^2 > 25$